sijantanv@gmail.com

Tanvir Ahmed Sijan

https://sijantanvir.github.io/

/tanvir-sijan

/sijantanvir

Research Interests

I am broadly interested in Computational Linguistics and Natural Language Processing, specifically (1) Human-AI alignment, (2) Developing robust evaluation frameworks for LLMs, and (3) Investigating model internals to advance LLM interpretability and AI safety.

Education

Jahangirnagar University — Dhaka, Bangladesh
Master of Science in Computer Science — *May 2025 – Sept. 2026*

- **Current CGPA**: 3.97/4.00
- **Relevant Coursework**: Machine Learning (A+), Deep Learning (A+), Natural Language Processing (A+), Data Mining (A), Artificial Intelligence & Expert Systems (A+), Computational Intelligence (A+)
- **CS Prerequisites**: Data Structures & Algorithms (A+), Software Engineering & Database Systems (A+), Computer Architecture & Operating Systems (A+), Programming Languages (A+)

Bangladesh University of Engineering and Technology (BUET) — Dhaka, Bangladesh
Bachelor of Science in Civil Engineering, with Honors — *Feb. 2020 – Mar. 2025*

- **CGPA**: 3.77/4.00, **Major**: Transportation Engineering, **Minor**: Environmental Engineering
- **Math and Programming Coursework**: Differential Equations & Statistics (A+), Differential, Integral Calculus and Matrices, Computer Programming Sessional (A+), Engineering Computation Sessional (A+)

Research Experience

Graduate Researcher — Jun. 2025 – Present
Computational Intelligence, Data and Language Research Lab — *Jahangirnagar University*
Supervisor: Dr. Md. Musfique Anwar

* Developed a Bangla event ontology, an annotated event detection dataset, and an evaluation framework for structured event trigger prediction to systematically evaluate the impact of annotation-guided instruction tuning and trigger- and context-level corruption on the performance and robustness of instruction-tuned generative LLMs and encoder-based models.
* Conceptualized and led the development of *BanglaSocialBench*, a culturally grounded benchmark spanning three sociopragmatic tasks for evaluating the cultural competence of large language models in Bangla.
* Contributed to the ideation, experimental design, and mechanistic analysis of the internal arithmetic circuits of LLMs by studying format-invariant arithmetic neurons across equivalent arithmetic problems reformulated as symbolic expressions, natural language word problems, and Python code.

Research Intern — May 2025 – Oct. 2025
Maks Inc. — *Dhaka, Bangladesh (HQ: Frisco, TX, USA)*

* Conducted literature reviews and collaborated with a team of data scientists and transportation engineers to analyze large-scale BRTA traffic datasets, develop emission models, perform geospatial analysis, conduct nationwide stakeholder surveys, and prepare technical reports for projects funded by the **World Bank** and **Asian Development Bank**.

Undergraduate Researcher — Nov. 2023 – Feb. 2025
Advised by: Dr. Md. Shamsul Hoque & Dr. Md Asif Raihan — *Bangladesh University of Engineering and Technology*

* Built an end-to-end computer vision and machine learning framework integrating street-view imagery, open geospatial data, and crash records to determine the impact of visual and built-environment characteristics on pedestrian crash severity, applying SHAP-based interpretability techniques.

Publications & Preprints

** indicates equal contribution*

[1] Sharath Naganna*, Tanvir Ahmed Sijan*, Uddipta Kalita. **"Are Arithmetic Heuristic Neurons Form-Invariant? A Mechanistic Analysis of Symbols, Text, and Code in LLMs."** *arXiv preprint arXiv:2607.16693*, July, 2026. [arXiv]

[2] Tanvir Ahmed Sijan, S. M Golam Rifat, Nayeemul Islam, Md. Musfique Anwar. **"Beyond Clean Text: Evaluating Encoder and Decoder Robustness for Bangla Event Detection in Noisy Text."** *arXiv preprint arXiv:2606.30914*, June, 2026. [arXiv]

[3] Tanvir Ahmed Sijan, S. M Golam Rifat, Pankaj Chowdhury Partha, Md. Tanjeed Islam, Md. Musfique Anwar. **"BanglaSocialBench: A Benchmark for Evaluating Sociopragmatic and Cultural Alignment of LLMs in Bangladeshi Social Interaction."** *ACL Student Research Workshop (Oral)*, 2026. **[Best Paper Award]** [ACL Anthology]

[4] Tanvir Ahmed Sijan, Md Asif Raihan, Sk. Md. Mashrur, Md. Shamsul Hoque. **"Assessing the Visual and Built Environment Risk on Pedestrian-Vehicle Crash Severity in Absence of Detailed Roadway Inventory: A Deep Learning Based Framework Harnessing Public Geospatial Data."** *Submitted to Transportation Research Board*, 2026.

Honors & Awards

Best Paper Award Jul. 2026
ACL Student Research Workshop 2026

Degree Awarded with Honors Mar. 2025
Bangladesh University of Engineering and Technology (BUET)

Undergraduate Student Research Grant Jan. 2024
Research and Innovation Centre for Science and Engineering, BUET

Dean's List Scholarship Jul. 2019
Bangladesh University of Engineering and Technology (BUET)
- Awarded for maintaining an average GPA of 3.75 or above across two consecutive regular terms.

3rd Place, Concrete Solutions Competition Mar. 2022
American Concrete Institute (ACI)
- Recognized for most innovative team design; featured in **Daily Prothom Alo** [Feature]

Higher Secondary Certificate (HSC) Scholarship 2019
Board of Intermediate and Secondary Education, Rajshahi
- Awarded **Talentpool Scholarship**; ranked **10th (∼top 0.01%)** among all candidates in the Rajshahi Board.

Service and Leadership

Workshop Reviewer Jul. 2026
11th Workshop on Natural User-generated Text (W-NUT) collocated with EMNLP 2026

Deputy Vice President Dec. 2024 – Mar. 2025
ITE BUET Student Chapter
- Collaborated with executive members to organize workshops and seminars on various topics of transportation research.

Vice President Mar. 2024 – Mar. 2025
ACI BUET Student Chapter
- Organized poster competitions, seminars, and field visits in coordination with chapter members.

Skills

Programming: Python, C, C++, SQL, MATLAB, LaTeX
Libraries: PyTorch, NumPy, Pandas, Transformers, Matplotlib, Scikit-Learn
Other: Linux, Git, Bash, LaTeX
Languages: Bangla (Native), English, Hindi (Basic), Arabic (Basic)